

Normal R chunks.

```
1 + 1

## [1] 2

x = rnorm(5)
```

Highlight matlab chunks. Note you need to install highlight from <http://www.andre-simon.de>, and you probably need to put its binary path into PATH; otherwise just use the engine.path option, e.g.
<<engine='highlight', engine.path = 'full/path/to/highlight'>>=

```
function Y = kalmanM(pos)
    dt=1;
    %% Initialize state transition matrix
    A=[ 1 0 dt 0 0 0;...      % [x ]
        0 1 0 dt 0 0;...      % [y ]
        0 0 1 0 dt 0;...      % [Vx]
        0 0 0 1 0 dt;...      % [Vy]
        0 0 0 0 1 0 ;...      % [Ax]
        0 0 0 0 0 1 ];        % [Ay]
    % Initialize measurement matrix
    H = [ 1 0 0 0 0 0; 0 1 0 0 0 0 ];
    Q = eye(6);
    R = 1000 * eye(2);
    x_est = zeros(6, 1);
    p_est = zeros(6, 6);

    numPts = size(pos,1);
    Y = zeros(numPts, 2);

    for idx = 1:numPts
        z = pos(idx, :)';

        %% Predicted state and covariance
        x_prd = A * x_est;
        p_prd = A * p_est * A' + Q;
        %% Estimation
        S = H * p_prd' * H' + R;
        B = H * p_prd';
        klm_gain = (S \ B)';
        %% Estimated state and covariance
        x_est = x_prd + klm_gain * (z - H * x_prd);
        p_est = p_prd - klm_gain * H * p_prd;
        %% Compute the estimated measurements
```

```
    Y(idx, :) = H * x_est;  
end % of the function  
end % of the function
```